RNS INSTITUTE OF TECHNOLOGY

Autonomous Institution Affiliated to VTU
RR Nagar, Bengaluru-560098

Estd : 2001

I / II Semester B.E. Degree Examination – June / July 2026
Python Programming

Time: 3 hrs **Max. Marks: 100**

Note:

1. ***Answer any 5 full questions, selecting at least one question from each module.***
2. ***Missing data may be suitably assumed.***
3. ***M: Marks, L: Bloom's level, C: Course Outcomes***

Q.No.		Module-1	M	L	C
Q1	a	Explain what a program is with a suitable example. Define debugging and describe the three types of program errors.	6	L2	CO1
	b	Explain break and continue statements with example code snippets.	6	L2	CO1
	c	Apply the concept of conditional execution in Python using if, elif, and else statements by writing a program that makes decisions based on given conditions. Demonstrate how these statements work with a suitable example.	8	L3	CO1
		OR			
Q2	a	What are functions? Explain Python function with parameters and return statements.	6	L2	CO1
	b	Write the differences between syntax, runtime, and semantic errors with examples.	6	L2	CO1
	c	Write a Python program to print all numbers from 1 to 20 using a for loop. Skip the numbers divisible by 3 and stop the loop if a number greater than 15 is encountered. Use range(), continue, and break statements appropriately.	8	L3	CO1
		Module-2			
Q3	a	Explain negative indexing, Slicing , index( ), append ( ), remove ( ), pop ( ), insert ( ), and sort ( ) with suitable example.	8	L2	CO2
	b	Develop a program to find mean, variance and standard deviation.	6	L3	CO2
	c	Develop a program to count the number of words in a given line of text.	6	L3	CO2
		OR			
Q4	a	Explain string as a compound data type. Demonstrate working with strings as single things using the upper() and lower() methods with examples.	8	L2	CO2
	b	Demonstrate how list mutability works by applying list operations to modify an existing list and compare the results with those obtained when creating and modifying a clone of the list. Illustrate your explanation with suitable examples.	6	L3	CO2
	c	Define a list. How is it different from an array? Develop a Python statement to access the third element of a list: nums = [3, 6, 9, 12].	6	L3	CO2
		Module-3			
Q5	a	Explain the process of writing to and reading from files using examples.	6	L2	CO2

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	b	Write a Python program to extract all email addresses from a text file using the findall() method.	8	L3	CO3
	c	What are Regular Expressions? Explain greedy and non-greedy matching with examples.	6	L2	CO3
		OR			
Q6	a	Explain get ( ) and setdefault ( ) method in dictionary.	6	L2	CO2
	b	What is a dictionary in Python? Explain aliasing and copying in dictionaries with examples.	6	L2	CO2
	c	Write a Python program that reads a file, counts the number of lines, words, and characters, and writes the result to another file.	8	L3	CO3
		Module-4			
Q7	a	What are classes and objects in Python.? Develop a python class named Employee with the attributes Name, ID, Department and Salary. Create a method work () and print the employee details.	10	L3	CO4
	b	How are namespaces, files and modules related? Explain three different ways to import names into the current namespace	10	L2	CO4
		OR			
Q8	a	Create a user-defined module in Python and demonstrate how it improves code reusability and readability by importing and using it in another program with suitable examples.	10	L3	CO4
	b	Explain the principles of Object-Oriented Programming in Python. Discuss classes, objects, attributes, methods, and scope with examples	10	L2	CO4
		Module-5			
Q9	a	What are the different types of errors that can occur in Python? What is the purpose of the try, except, else, and finally blocks? Write a Python program that handles invalid input using try and except blocks.	10	L3	CO5
	b	Explain the concept of generalization in inheritance. How does it help in reducing code duplication? Support your answer with a class hierarchy example.	10	L2	CO5
		OR			
Q10	a	Write short notes on raising custom exceptions in Python. Discuss its importance in robust program design.	10	L2	CO5
	b	Write a Python program to demonstrate operator overloading by overloading the + operator in a user-defined class	10	L3	CO5

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CIE – Test 1

1	R	N	2	5	C	D	0		

Sem: II
Subject: Python Programming
Subject Code: BPRLC205B

Date: 17.04.2026
Time: 9:30 am – 11:00 am
Max. Marks: 50

Instruction: Answer any 5 full questions, selecting ONE question from each part.

Q.No.		Question	Marks	RBT*	COs
1	a	What are the rules for naming a variable? State whether the following variable names are valid/invalid? Justify your answer. i)My-var ii) _2var iii) for iv) false v) print vi) my var	5	L2	CO1
	b	Develop a Python program to find the sum of the digits of a given integer number. *[Note: Do not convert the integer into a string; process the number using arithmetic operations only.]*	5	L3	
		OR			
2	a	Develop a Python program that assigns an integer to the variable named status and then: prints "Stopped" if status == 0, prints "Running" if status == 1, prints "Unknown status" for any other value.	5	L2	CO1
	b	Develop a program that uses while loop to add up all the odd numbers between 3 and 300, inclusive, and displays the result.	5	L3	
3	a	Explain break and continue statements with example code snippets.	5	L2	CO1
	b	Develop a function which looks through a list of words. It should return the first 2-letter word. If there is no 2-letter word, it should return the empty string.	5	L3	
		OR			
4	a	Explain the purpose and working of the following list methods in Python with suitable examples and output: append(), insert(), count(), extend(), index(). Describe how each method affects the list.	5	L2	CO1
	b	Develop a Python program that defines a function **is_prime(n)** which returns True if n is prime and False otherwise, then uses that function to read a number and print whether the number is prime.	5	L3	
5	a	What do the following expressions evaluate to? i) 'Pythonic-Guide'[1] ii) 'Pythonic-Guide'[0:6] iii)'Pythonic-Guide'[:7] iv)'Pythonic-Guide'[8:] v) 'Pythonic-Guide'[:50]	5	L3	CO2
	b	Develop a Python function **analyse(record)** that accepts a tuple	5	L3	

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		containing a student's name followed by 5 subject marks, and returns the total, average, highest, and lowest marks as a tuple.			
		OR			
6	a	Develop a Python program to read N numbers from the console and create a list. Also, print the mean, variance and standard deviation with suitable messages.	5	L3	CO2
	b	Develop a Python function count_vowels(tuple) that accepts a tuple of characters and returns the number of vowels and consonants as a tuple.	5	L3	
7	a	Develop a Python program to find the second largest element in a list of n integers entered by the user. Handle the case where duplicate values are present.	5	L3	CO2
	b	Develop a program to delete all occurrences of a given particular value from a given list. Read the list and the item to be removed from the user.	5	L3	
		OR			
8	a	What will be the output of the following code? Justify your answer. a = "One/Two/Three" s = a.split("/") s1 = "-".join(s) print(s) print(s1)	5	L3	CO2
	b	Read a multi-digit number (as chars) from the console. Develop a program to print the frequency of each digit with a suitable message.	5	L3	
9	a	What is a dictionary in Python? Explain how it differs from a list with suitable examples.	4	L2	CO3
	b	Develop a Python program that reads a multi-digit integer from the user and counts the frequency of each digit using a dictionary. *[Note: Do not convert the integer into a string; process the number using arithmetic operations only.]*	6	L3	
		OR			
10	a	Explain Dictionary copying and aliasing with suitable examples.	4	L2	CO3
	b	Develop a Python program to create an empty dictionary and use setdefault() to add the keys "city" and "country" with appropriate values. If we try adding "city" again with a different value using setdefault(), what will be the result. Justify your answer.	6	L3	

*** Revised Bloom's Taxonomy: L1-Remember, L2-Understand, L3-Apply, L4-Analyze, L5-Evaluate, L6-Create**

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CIE – Test 2

1	R	N	2	5	C	D	0		

Sem: II
Subject: Python Programming
Subject Code: BPRLC205B

Date: 02.06.2026
Time: 9:30 am – 11:00 am
Max. Marks: 50

Instruction: Answer any 5 full questions, selecting ONE question from each part.

Q.No.		Question	Marks	RBT*	COs
1		Define Directories.Develop a program to display contents of a folder recursively (Directory) having sub-folders and files (name and type).	10	L3	CO3
		OR			
2		Illustrate the use of Regular Expressions in pattern matching for text processing with suitable syntax, key functions, and examples.	10	L3	CO3
3		Develop a python program to create a class called time with members hours, minutes, and seconds.Store user-entered time and print the total number of seconds.	10	L3	CO4
		OR			
4	a	Discuss different forms of the **import** statement and explain their syntax with examples.	5	L3	CO4
	b	Develop a Python program to define a class Student with attributes name and marks. Include a method to display details.	5	L3	
5		List the principles of Object-Oriented Programming in Python and explain classes, objects, attributes, methods, and scope with an suitable examples.	10	L2	CO4
		OR			
6	a	Define a class. Explain attributes and methods with an example class.	5	L2	CO4
	b	What is a namespace? Explain scope and lookup rules in Python.	5	L2	
7		Define exceptions in Python. Explain the four clauses used in exception handling and describe how custom exceptions can be created and raised in Python with suitable examples.	10	L2	CO5
		OR			
8	a	What is inheritance? Explain with a suitable example program.	5	L2	CO5
	b	Explain pure functions and modifiers in the context of object-oriented programming.	5	L2	
9		Develop a function named DivExp which takes TWO parameters a, b, and returns a value c (c = a/b).Write a suitable assertion for a > 0 in the function DivExp and raise an exception for when b = 0.Develop a suitable program that reads two console values and calls the function DivExp.	10	L3	CO5
		OR			

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10	a	What is the output of the following Python program? Justify your answer by explaining the role of __add__(), __mul__(), and __str__() methods.
		class Box:
		
		def __init__(self,l,b):
		self.l=l
		self.b=b
		
		def __add__(self,other):
		return Box(self.l+other.l,self.b+other.b)
		
		def __mul__(self,other):
		return Box(self.l*other,self.b*other)
		
		def area(self):
		return self.l*self.b
		
		def __str__(self):
		return "({0},{1})".format(self.l,self.b)
		
		b1 = Box(2,5)
		b2 = Box(4,3)
		
		b3 = b1 + b2
		b4 = b3 * 2
		
		print(b3)
		print(b4)
		print(b4.area())
			6	L3	CO5

	b	Make use of a suitable example to explain the operator overloading concept in python.	4	L3	

*** Revised Bloom's Taxonomy: L1-Remember, L2-Understand, L3-Apply, L4-Analyze, L5-Evaluate, L6-Create**

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CIE – Test 2

Sem: I
Subject: Programming in C
Subject Code: BCSEN103

Date: 29-12-2025
Time: 9:30-11:00
Max. Marks: 50

Instruction: Answer any 5 full questions, selecting ONE question from each part.

Q.No.		Question	Marks	RBT*	COs
1	a	Define function and explain the categories of user defined functions.	6	L2	CO4
	b	Develop a C program to find factorial of a given number using Recursion.	4	L3	
		OR			
2	a	Explain the elements of user defined functions with examples.	6	L2	CO4
	b	Demonstrate parameter passing techniques in C using suitable code snippet.	4	L3	
3	a	What is pointer? Explain how the pointer variable declared and initialized?	5	L2	CO4
	b	Develop a program to find GCD and LCM of two numbers using concept of functions.	5	L3	
		OR			
4	a	Explain the array of pointers with example? or explain how pointers and arrays are related with example	5	L2	CO4
	b	Develop a C program to find the sum and mean of all elements in an array using pointer.	5	L3	
5	a	Demonstrate a C program to illustrate structures to read and write Book-Title, Book-Author and Book-id of n books.	10	L3	CO3
		OR			
6	a	Demonstrate with an algorithm and program to read N integers and to arrange them in ascending order using bubble sort technique.	10	L3	CO3
7	a	Implement a C program to search an element using binary search techniques.	6	L3	CO5
	b	Explain enumerated data types with example.	4	L2	
		OR			
8	a	Implement a C program that accepts a structure variable as a parameters to a function from a function call.	6	L3	CO5
	b	Explain the differences between structures and unions.	4	L2	
9	a	Develop a C program to maintain a record of n students details using an	6	L3	CO

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		array of structures with four fields-Roll number, name, marks and grade. Calculate the grade according to the following conditions. Marks grade >=80 A ,>=60 B,>=50 C,>=40 D			
	b	Write a note on Bit-Fileds.	4	L2	
		OR			
10	a	Demonstrate array of structures and structure within structure with example.	6	L3	CO5
	b	What is preprocessor directive ? Explain any two preprocessor directive	4	L2	

*** Revised Bloom's Taxonomy: L1-Remember, L2-Understand, L3-Apply, L4-Analyze, L5-Evaluate, L6- Create**

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